

MEDCORE HMS

Hospital Management System

Complete Build Roadmap — Gap-Filled & Competitive

*An ABDM-native, compliance-first, 100% free-and-open-source stack
designed for small & mid-size Indian clinics and hospitals*

Team size: 2 developers Realistic timeline: ~12 months (pilot-ready MVP by month 4)

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1. What Changed From the Original Roadmap

The original 8-month plan was well-structured but had gaps that would have been expensive to fix later, and it did not yet account for selling into a crowded, compliance-driven market. This version fills every gap identified and reorganises the work so that a sellable product reaches a pilot hospital early, then grows into a full hospital system. The core stack stays free and open-source; the changes are about sequencing, compliance, and competitive positioning.

Gap in original plan	How this roadmap fixes it
No ABDM / ABHA / FHIR support	Schema is FHIR-shaped and audit-ready from Phase 1; a dedicated ABDM integration phase (Phase 6) builds ABHA creation, consent, and FHIR record exchange on the free NHA sandbox.
Outpatient-only (a CMS, not an HMS)	A full IPD module (Phase 8) adds wards, beds, admissions/ADT, and nursing — so the product can be sold to hospitals with beds, not only clinics.
create-react-app (deprecated)	Frontend built on Vite; mobile on Expo/React Native. Both free, both current.
Vague 'offline' promise	Offline is scoped explicitly: LAN-first architecture plus an optional local read cache; true multi-site sync is deferred and labelled as such.
Testing stacked at the very end	Automated tests are written alongside every phase; CI runs them on every commit from Phase 1.
DPDP & audit logs treated as notes	Security, RBAC, audit logging, and encryption are a first-class foundation built in Phase 1, not bolted on.
Single-tenant maintenance ignored	Phase 10 delivers one-command deployment, central monitoring, and automated backups so two people can support many installs.
No competitive wedge	Section 3 defines a narrow, defensible positioning; differentiation features are built into Phase 9.
Unrealistic solo timeline	Re-scoped to ~12 months for a team of 2, with an early pilot-ready MVP milestone at ~month 4.

2. Positioning — How This Competes and Wins

The Indian HMS market is large and growing, but crowded with mature vendors. A two-person team cannot win a feature war against companies with sales teams and hundreds of reference hospitals. It can win by being the obvious best choice for a narrow slice, and by turning free-and-open-source into a genuine price advantage. This roadmap is built around four defensible edges.

Edge 1 — Radically lower total cost of ownership

Because every component is free and open-source, there are no per-seat licence fees to pass on. Incumbents priced from roughly ₹40K/month become hard to justify for a small facility when a comparable, ABDM-ready system runs on a modest server with only a support subscription. Free tooling is the product's structural advantage — protect it.

Edge 2 — ABDM-native, not ABDM-retrofitted

Many legacy systems are struggling to bolt ABDM onto old data models. Building FHIR-shaped data and a consent workflow in from the start means the product is compliant by design — a concrete selling point to any AB-PMJAY-empanelled hospital that risks de-empanelment without it.

Edge 3 — One specialty, done perfectly (recommended focus)

Rather than a generic system for everyone, pick a single facility type for launch — for example single-specialty clinics (dental, eye, or dermatology) or 20–50 bed nursing homes. Ship workflows, templates, and reports so tailored that a general HMS feels clumsy by comparison. Widen later once you have reference customers. The narrower the launch target, the stronger a small team's advantage of focus and speed.

Edge 4 — Local presence, service, and language

Big vendors give small-town hospitals poor support. Being local, showing up in person, and supporting the regional language (built in Phase 9) is something remote incumbents cannot match cheaply.

Do this first: Before Phase 1 hardening begins, complete Phase 0 discovery — talk to 5–10 target facilities, confirm the launch specialty, and sign at least one design-partner. The roadmap assumes this validation runs in parallel with the earliest setup work.

3. Complete Free & Open-Source Tool Stack

Every tool below is free to use. Where a paid element is genuinely unavoidable (payment-gateway transaction fees, a purchased domain, optional cloud backup storage), it is called out explicitly. Nothing here requires a commercial licence.

3.1 Core platform & backend

Tool	Role	Licence / cost
Ubuntu Server 24.04 LTS	Hospital server OS	Free
Python 3.12	Backend language	Free
FastAPI + Uvicorn	REST API framework & ASGI server	Free
PostgreSQL 16	Primary relational database	Free
SQLAlchemy + Alembic	ORM and schema migrations	Free
Pydantic	Validation / schemas	Free
Redis	Caching, rate-limit, background-job queue	Free
Celery	Async tasks (alerts, report generation)	Free

3.2 Security, auth & compliance

Tool	Role	Licence / cost
Keycloak	Identity, RBAC, SSO, MFA (open-source auth server)	Free
Argon2 / passlib	Password hashing	Free
python-jose	JWT tokens	Free
PostgreSQL TDE / pgcrypto + LUKS	Encryption at rest (disk + column)	Free
Caddy	Reverse proxy with automatic HTTPS (Let's Encrypt)	Free
HAPI FHIR (optional)	Standalone FHIR R4 server / validator	Free
LOINC, ICD-10, SNOMED CT (India)	Clinical terminologies (India has free NRCeS access)	Free*

* SNOMED CT is free to use in India through the National Release Centre (NRCeS); confirm current registration terms before shipping.

3.3 Frontend, mobile & desktop

Tool	Role	Licence / cost
React + Vite	Staff web application (replaces CRA)	Free
TypeScript	Type-safe frontend	Free
Ant Design / shadcn-ui	UI component library	Free
TanStack Query	API data fetching & caching	Free
React Router, dayjs, Recharts	Routing, dates, charts	Free
i18next	Multi-language (regional language support)	Free
Expo + React Native	Patient mobile app (Android/iOS)	Free
Tauri	Desktop wrapper / installable .exe	Free

3.4 DevOps, testing, monitoring & backups

Tool	Role	Licence / cost
Git + GitHub (free private repos)	Version control	Free
Docker + Docker Compose	Reproducible packaging & one-command install	Free
GitHub Actions	CI/CD (build, test, release)	Free tier
pytest + httpx	Backend unit & integration tests	Free
Playwright	End-to-end frontend tests	Free

Tool	Role	Licence / cost
Locust	Load testing	Free
Prometheus + Grafana	Metrics & dashboards	Free
Loki	Centralised log aggregation	Free
Uptime Kuma	Per-install uptime monitoring	Free
GlitchTip (self-hosted Sentry)	Error tracking	Free
pg_dump + cron + rclone	Automated encrypted nightly backups	Free**

** Backup software is free; off-site cloud storage may cost a small amount per hospital, or use the hospital's own second disk / NAS at no cost.

3.5 Documents, design & payments

Tool	Role	Licence / cost
WeasyPrint / ReportLab	PDF bills, receipts, prescriptions (A4 + thermal)	Free
openpyxl + pandas	Excel exports & reports	Free
Penpot / Figma (free tier)	UI/UX design	Free
Razorpay / UPI SDK	Digital payments (optional)	Free SDK***

*** The integration SDK is free; payment gateways charge a per-transaction fee. Cash/UPI-QR collection has no software cost.

4. Compliance Foundation (Built From Day One)

These are not features to add if time allows — for Indian healthcare software they are the price of entry and a legal responsibility once the product is sold. They are established in Phase 1 and enforced throughout.

4.1 DPDP Act readiness

- Encryption at rest (disk-level LUKS + column-level pgcrypto for sensitive fields) and in transit (TLS everywhere via Caddy).
- Role-based access control: no user sees data outside their role; enforced at the API layer, not just the UI.
- Data-subject rights: ability to export and, where lawful, erase a patient's data; consent capture recorded with timestamp.
- Data residency: single-tenant on the hospital's own server satisfies the common 'our data stays with us' requirement.

4.2 Audit logging (first-class)

A dedicated, append-only audit_log table plus API middleware records who accessed or changed which patient record, when, and from where. This is mandatory for healthcare and is a frequent question during hospital sales. It cannot be added convincingly later, so it ships in Phase 1.

4.3 ABDM / FHIR readiness

- Clinical data (diagnoses, prescriptions, lab results) is stored with coded terminologies (ICD-10 / SNOMED / LOINC) so it can be emitted as FHIR R4 bundles.
- The patient model carries optional ABHA number and ABHA address fields from the start.
- Full ABHA creation, consent management, and Health Information Provider (HIP) record exchange are built in Phase 6 against the free ABDM sandbox, then certified for production.

5. Revised Database Schema (FHIR-Shaped, Audit-Ready, IPD-Capable)

The schema below extends the original with compliance, ABDM, and inpatient tables. Coded fields replace free text where FHIR needs them. This is designed so that ABDM and IPD do not require a rewrite later.

Core & identity

- patients — patient_id, uhid, abha_number (nullable), abha_address (nullable), name, dob, gender, phone, address, blood_group, consent_status, created_at
- users — user_id, name, role, keycloak_id, status, created_at
- doctors — doctor_id, user_id, name, specialization, hpr_id (ABDM provider id), department_id, phone
- departments — dept_id, name, type (OPD/IPD/Lab/Pharmacy/Radiology)
- facility — facility_id, name, hfr_id (ABDM facility id), address, gstin

Clinical (coded for FHIR)

- appointments — appt_id, patient_id, doctor_id, date_time, status, queue_no
- consultations — consult_id, patient_id, doctor_id, chief_complaint, diagnosis_code (ICD-10), diagnosis_text, notes, date
- lab_orders — order_id, consult_id, test_code (LOINC), test_name, status, ordered_by, ordered_at
- lab_results — result_id, order_id, result_data, reference_range, uploaded_by, uploaded_at
- prescriptions — rx_id, consult_id, instructions, created_at
- prescription_items — item_id, rx_id, med_code, med_id, dose, frequency, duration

Pharmacy & inventory

- medicines — med_id, name, code, category, unit, hsn_code, gst_rate, unit_price, reorder_threshold
- stock_batches — batch_id, med_id, batch_no, expiry_date, quantity, purchase_price
- pharmacy_dispensing — dispense_id, rx_id, med_id, batch_id, quantity, dispensed_by, dispensed_at

Billing

- bills — bill_id, patient_id, visit_id, consult_fees, lab_fees, pharmacy_fees, room_charges, gst_total, total, status, payment_mode, paid_at
- bill_items — line_id, bill_id, description, hsn_code, qty, rate, gst_rate, amount

Inpatient / IPD (new)

- wards — ward_id, name, type, floor
- beds — bed_id, ward_id, bed_no, status (vacant/occupied/maintenance), daily_rate
- admissions — admission_id, patient_id, bed_id, admitting_doctor_id, admit_time, discharge_time, status (ADT lifecycle)
- nursing_notes — note_id, admission_id, vitals (JSON), note_text, recorded_by, recorded_at

Compliance & sync

- audit_log — log_id, actor_user_id, action, entity, entity_id, ip_address, timestamp (append-only)
- consent_artifacts — consent_id, patient_id, purpose, granted_at, expiry, abdm_consent_ref
- fhir_exports — export_id, entity, entity_id, bundle_ref, status, created_at

6. Phase-by-Phase Roadmap

Twelve phases across ~52 weeks for two developers. The plan is deliberately MVP-first: a sellable clinic system reaches a pilot at roughly month 4, then matures into a full hospital platform. Each phase lists its goal, the free tools it uses, the key work, and an exit criterion that must be met before moving on.

Phase 0: Discovery & Validation (Weeks 1–2, ongoing)

Goal — Confirm who you are selling to and lock the launch specialty before heavy building.

Free tools: Notebook, phone, Penpot for mockups.

- Interview 5–10 target facilities: what they use today, what they hate, whether AB-PMJAY empanelled, willingness to pay.
- Choose ONE launch facility type (e.g. single-specialty clinic) and sign at least one design-partner.
- Draft low-fidelity mockups and validate the core screens with the design-partner.

Exit criterion: At least one facility has verbally committed to pilot the product.

Phase 1: Foundation & Compliance Scaffolding (Weeks 1–4)

Goal — Stand up the environment, the FHIR-shaped schema, and the security/audit foundation.

Free tools: Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, Keycloak, Docker Compose, GitHub, GitHub Actions, Caddy.

- Repo, branching model, and CI pipeline running on the first commit (lint + tests + build).
- Full schema (Section 5) as Alembic migrations, including audit_log, consent_artifacts, IPD tables.
- Keycloak configured with roles (admin/doctor/nurse/lab/pharmacy/billing); JWT validation middleware.
- Audit-logging middleware and encryption (TLS via Caddy, pgcrypto for sensitive columns).
- Dockerised local dev stack (API + DB + Redis + Keycloak) that starts with one command.

Exit criterion: A new developer can clone, run `docker compose up`, log in with a role, and see an audit entry created.

Phase 2: Core Backend API + Continuous Testing (Weeks 4–8)

Goal — Build and test the core clinical and administrative endpoints.

Free tools: FastAPI, Pydantic, pytest, httpx, Redis, Celery.

- Endpoints: auth, patients (register/search/update), doctors & schedules, consultations, lab orders/results, prescriptions.
- Coded-field validation (ICD-10 / LOINC) so records are FHIR-ready.
- pytest unit + integration tests written alongside each endpoint; CI blocks merges on test failure.
- Queue/appointment logic and background tasks (Celery) for notifications and stock checks.

Exit criterion: Full OPD workflow passes an automated end-to-end integration test.

Phase 3: Staff Web Application (MVP) (Weeks 8–12)

Goal — Deliver the daily-use staff app that reaches the pilot.

Free tools: React + Vite, TypeScript, Ant Design, TanStack Query, Recharts, dayjs.

- Role-based login and dashboards (admin, doctor, lab, pharmacy, billing).
- Screens: patient registration & search, doctor consultation, lab module, prescription view/print, basic billing.
- Playwright end-to-end tests for the critical screens.
- Print templates (A4 + thermal) via the PDF service.

Exit criterion: ★ MVP MILESTONE (~Month 4): the design-partner clinic can run a full patient visit end to end.

Phase 4: Pharmacy & Inventory (Weeks 12–16)

Goal — Full pharmacy module with batch/expiry tracking and stock alerts.

Free tools: FastAPI, APScheduler/Celery, openpyxl, WeasyPrint.

- Medicine master with HSN/GST; batch and expiry management; automatic stock deduction on dispense.
- Low-stock and near-expiry alerts; full dispensing audit trail.
- Stock reports to Excel; feed of dispensed costs into billing.

Exit criterion: Dispensing against a prescription correctly deducts the right batch and raises a low-stock alert.

Phase 5: Billing, Payments & Reports (Weeks 16–20)

Goal — Aggregate charges, collect payment, and produce GST-compliant documents.

Free tools: WeasyPrint/ReportLab, pandas, openpyxl, Razorpay SDK (optional), UPI-QR.

- Auto-aggregate consultation + lab + pharmacy (+ room, later) into an itemised, GST-correct bill.
- Payment collection (cash, UPI-QR, optional card/UPI gateway); PDF receipts (A4 + thermal).
- Daily/monthly revenue and outstanding reports for admin.

Exit criterion: A patient visit produces a correct GST bill and receipt; daily revenue report reconciles.

Phase 6: ABDM / ABHA / FHIR Integration (Weeks 20–26)

Goal — Make the product ABDM-compliant — the key competitive and regulatory edge.

Free tools: ABDM sandbox (free), HAPI FHIR validator, fhir.resources (Python).

- Register facility (HFR) and providers (HPR) in the sandbox; ABHA creation/verification at registration (Milestone 1).
- Emit consultations, prescriptions, lab reports, discharge summaries as FHIR R4 bundles linked to ABHA (Milestone 2).
- Consent Manager flow: HIP record sharing with patient consent (Milestone 3).
- Validate all bundles against the FHIR profile; then apply for production certification.

Exit criterion: M1–M2 flows pass in the ABDM sandbox and bundles validate against the India FHIR profile.

Phase 7: Patient Portal (Web + Mobile App) (Weeks 26–32)

Goal — Give patients secure access to THEIR OWN records — the doctor's notes, no AI-generated medical content.

Free tools: React + Vite (portal), Expo + React Native (app), i18next.

- Secure patient login (OTP/ABHA); view own diagnoses, prescriptions, lab results, and visit history.
- Doctor-written care instructions shown in plain language and regional language; downloadable/printable.
- Optional curated, doctor-attached educational material (a library — never free-form or AI-generated).
- Appointment booking and bill/receipt access.

Exit criterion: A patient logs in on web and app and sees only their own doctor-authored records and instructions.

Phase 8: Inpatient (IPD) Module (Weeks 32–38)

Goal — Turn the clinic system into a true hospital system with beds and admissions.

Free tools: FastAPI, React + Vite.

- Ward & bed management with live occupancy board; admission/discharge/transfer (ADT) lifecycle.
- Nursing notes and vitals capture; room charges flowing into billing automatically.
- Discharge summary generation (also emitted as a FHIR bundle).

Exit criterion: A patient can be admitted, moved between beds, charged room fees, and discharged with a summary.

Phase 9: Analytics, Differentiation & Localisation (Weeks 38–42)

Goal — Build the features that make the product feel better than a generic HMS.

Free tools: Recharts/Grafana embeds, i18next, specialty templates.

- Admin analytics: patient flow, revenue, doctor productivity, inventory turns.
- Specialty-specific templates and workflows for the chosen launch vertical.
- Full regional-language UI; thermal + A4 print polish; configurable branding per hospital.

Exit criterion: The launch-specialty workflow is measurably faster than a generic flow in a timed test with the design-partner.

Phase 10: Deployment Automation, Monitoring & Hardening (Weeks 42–46)

Goal — Make single-tenant installs cheap to deploy and safe to run — the operational make-or-break.

Free tools: Docker Compose, Caddy, systemd, Prometheus + Grafana, Loki, Uptime Kuma, GlitchTip, pg_dump + cron + rclone, Tauri.

- One-command, identical install script per hospital (same OS, same containers).
- Automated encrypted nightly backups with tested restore; pre-update backup + rollback.
- Central monitoring: uptime, errors, disk, and backup-success alerts across all installs.
- Security pass: OWASP checklist, dependency scanning, penetration self-test; package the Tauri desktop build.
- Update mechanism: admin-approved, versioned, with automatic backup before applying.

Exit criterion: A fresh hospital server goes from bare Ubuntu to a running, monitored, backed-up install in under an hour.

Phase 11: Pilot Rollout, Feedback & Launch (Weeks 46–52)

Goal — Prove it in the real world, harden, and turn the pilot into a reference customer.

Free tools: Locust (load), Playwright (regression), all monitoring tools.

- Full install at the design-partner; staff training and an installation checklist document.
- Load and security testing; fix issues surfaced by real use.
- Collect testimonials/metrics; convert the pilot into a paying reference customer.
- Prepare the sales collateral (ABDM-ready, DPDP-compliant, TCO comparison).

Exit criterion: The pilot hospital is live, paying, and willing to be a reference for the next sale.

7. Deployment & Operations (Single-Tenant, Sustainable for Two People)

Single-tenant means every hospital runs its own copy — good for data-residency sales, but a maintenance multiplier. These practices keep the support load survivable for a two-person team.

- Identical installs: every deployment is the same Docker Compose stack on the same Ubuntu LTS, so there is one configuration to reason about, not N.
- One-command provisioning: a single script installs, configures HTTPS (Caddy), seeds the database, and registers the install with central monitoring.
- Central monitoring: Prometheus/Grafana + Uptime Kuma + GlitchTip give one dashboard across all hospitals, so you learn about problems before the 2am phone call.
- Automated, tested backups: encrypted nightly pg_dump, off-site copy, and — critically — a scheduled restore test so backups are known-good.
- Safe updates: versioned releases, admin-approved, automatic pre-update backup, and a one-command rollback.
- Capacity honesty: two people can sustainably support a limited number of installs; plan growth around that, or move heavier hospitals toward a managed/hybrid hosting option later.

Deployment models to revisit at scale: Single-tenant on-prem is right for launch and for data-residency-sensitive buyers. Once you have several customers, consider an optional managed-cloud (still your software, you host it) tier — it collapses N installs into a few, dramatically reducing support load. Keep the on-prem option for those who require it.

8. Business & Pricing Model

The software being free to build does not mean it should be free to buy — the value is in the working system, the compliance, and the support. Price for a sustainable two-person operation.

- Recurring, not one-time: charge an annual (or monthly) subscription / AMC that covers updates, support, and backups. A one-time licence with free support will bury you.
- Tiered by size: price by number of doctors or beds so a small clinic pays little and a mid-size hospital pays more, matching their ability to pay.
- Implementation fee: a one-time setup/training/data-migration fee for each new install compensates the real work of onboarding.
- Undercut on TCO, not on service: your price advantage comes from zero licence cost — lead with total-cost-of-ownership vs. incumbents, while charging enough to fund good support.
- Support tiers: a basic tier (business hours) and a premium tier (extended/priority) let hospitals self-select and cap your obligation.

9. Timeline Summary

Phase	Weeks	Month	Key deliverable
0 Discovery & validation	1–2 (ongoing)	1	Signed design-partner; launch specialty chosen
1 Foundation & compliance	1–4	1	Schema, security, audit, CI running
2 Core backend + tests	4–8	2	OPD API passing end-to-end tests
3 Staff web app (MVP)	8–12	3–4	★ Pilot-ready clinic system
4 Pharmacy & inventory	12–16	4–5	Batch/expiry stock + dispensing
5 Billing & payments	16–20	5–6	GST bills, receipts, reports
6 ABDM / FHIR	20–26	6–7	Sandbox M1–M3; FHIR bundles valid
7 Patient portal + app	26–32	7–8	Patients see own records (web + app)
8 IPD module	32–38	8–9	Wards, beds, ADT, nursing
9 Analytics & localisation	38–42	9–10	Specialty templates, regional language
10 Deploy, monitor, harden	42–46	10–11	One-hour install; central monitoring
11 Pilot rollout & launch	46–52	11–12	Live, paying reference customer

Sequencing note: Phases 4 and 5 can overlap, and ABDM (6) can begin in parallel once the schema is stable. The MVP at month 4 exists specifically so you can start earning trust and feedback long before the full platform is finished.

10. Key Risks & Mitigations

Risk	Mitigation
Build the wrong thing	Phase 0 validation + early MVP pilot; let the design-partner steer features.
ABDM certification takes longer than expected	Start sandbox work early (parallel to Phase 4–5); consider a free/low-cost integration layer if direct certification stalls.
Support load exceeds two people	Identical installs, central monitoring, automated backups; plan a managed-hosting tier before scaling install count.
Data-loss incident (reputational + legal)	Encrypted, tested backups; rollback on every update; DPDP controls from Phase 1.
Incumbents undercut or out-feature you	Compete on TCO + ABDM-native + one specialty done perfectly + local service — not on breadth.
Scope creep delays launch	Protect the month-4 MVP milestone; everything after it is additive, not blocking.

This roadmap is a technical and go-to-market plan, not legal or financial advice. ABDM, AB-PMJAY, DPDP, and GST requirements evolve — verify current NHA/NHA-sandbox certification steps and compliance obligations before production launch. Market and tooling details drawn from 2025–2026 sources should be re-confirmed.